

ZHOU Ziyang

Phone: (86) 15757817693 | Email: zhouziyang20050630@163.com

Personal Website: <https://ziyangzhou.me/> (portfolio & project archive)

EDUCATIONAL BACKGROUND

University of British Columbia

09/2023 – 05/2027

Bachelor of Science, Major in Statistics program

British Columbia, Canada

Core Modules: Differential Calculus with Applications (88), Computation, Programs, and Programming (86), Basic Algorithms and Data Structures (86), Applied Machine Learning (83), Models of Computation (81), Calculus III (84), Calculus IV (81), Elementary Statistics for Applications (80), Finding Relationships in Data (80)

Honors: Dean's List, Year 1 (2023 – 24 Winter Session)

UBC, Canada

WORK EXPERIENCE

Anhui Shusheng Data Technology Co., Ltd.

Hefei, China

Data Analyst Intern

03/08/2026 – 31/08/2026 & 06/05/2026 – 31/05/2026

- Independently collected, integrated, and standardized multi-source power system data covering historical electricity sales, load, meteorological conditions, holidays, and user attributes, processing daily data volumes at the thousand-record level and constructing high-quality modelling datasets for subsequent forecasting analysis
- Participated in data processing and parameter tuning for weekly and monthly electricity consumption forecasting models, extracting and selecting time-series features (trend, seasonality) and external features (temperature, holidays, user capacity, industry growth cycles)
- Reduced an initial set of 20+ candidate features to 12 core features through correlation analysis, and built and tuned sub-models, including Random Forest, to capture non-linear relationships among features
- Ensembled multiple sub-models using weighted averaging, with weights iteratively adjusted through backtesting on historical data, contributing to improved forecasting accuracy and robustness compared to single-model approaches
- Compiled prediction results and generated data visualizations for project reporting, comparing model outputs across configurations and supporting client-facing forecast presentations

Shanghai Guangbo Hi-intelligence Technology Co., Ltd.

Shanghai, China

Actuarial Department Intern

08/06/2026 – 31/07/2026

- Cleaned, integrated, and standardized client business data using SAS, covering policy records across multiple product lines with daily processing volumes at the thousand-record level, supporting the data foundation for actuarial modelling
- Participated in the application of Generalized Linear Models (GLM) to insurance pricing, assisting in the screening, validation, and coefficient calibration of 15+ risk factors, and comparing model outputs across different factor specifications to support pricing decisions
- Independently extracted and verified rate table factors, and co-authored the Standard Operating Manual for Rate Table Extraction, reducing the time spent on repetitive manual verification tasks by approximately 25% through workflow abstraction, establishing a reusable template for future pricing cycles
- Selected and curated daily industry intelligence, distributing it to team members and cross-functional colleagues, supporting team-wide awareness of regulatory updates and market trends

PROJECT EXPERIENCE

Tencent AI-HR 90-Day Onboarding System, Independent Developer

06/2026

Tech Stack: Multi-Agent Collaboration, Codex + CodeBuddy, AI-assisted Deployment, Database Integration

- Designed a 90-day onboarding pathway prototype, integrating new hire profiling, growth task tracking, mentor feedback, and HR oversight into a cohesive AI-HR use case
- Implemented a Codex-CodeBuddy collaborative workflow, enabling different AI tools to delegate tasks across agents for implementation, deployment, debugging, logic review, and language refinement
- Leveraged AI-assisted coding for web deployment, database connectivity, and troubleshooting, while manually validating workflow logic, data persistence, error recovery, and demo stability to ensure end-to-end viability

Local Qwen Chinese Short Story Generation Pipeline, Independent Developer

05/2026 – 08/2026

Tech Stack: Local Qwen Model, Python, Agent Memory Mechanism, Prompt Engineering

- Built a local Qwen-based generation pipeline for automated short story creation using Python, decomposing the writing

process into modular stages including world setting, outline planning, and chapter-wise continuation, simulating an agent-driven, self-sustaining writing workflow

- Implemented a lightweight agent memory mechanism that retrieves prior settings, fragments, and logs to maintain character consistency and plot coherence across chapters, addressing common challenges in long-form text generation
- Designed a self-questioning and self-answering workflow that orchestrates model invocation and iterative refinement, enabling the model to generate questions and answers autonomously to sustain narrative progression

NLP Text Analysis & Topic Modeling, Independent Developer

01/2026 – 04/2026

Tech Stack: Python, Jupyter Notebook, Scikit-learn, spaCy, Gensim, GloVe, LDA

- Built end-to-end NLP pipelines spanning text preprocessing, word embedding exploration, and topic modeling, delivering interpretable results from raw text corpora
- Applied GloVe embeddings to analyze word similarity, nearest neighbors, vocabulary coverage, and representation biases, assessing the limitations of pre-trained embeddings on domain-specific texts
- Constructed topic models using CountVectorizer and Latent Dirichlet Allocation (LDA), examining word-topic and document-topic distributions to interpret thematic structures across diverse text collections

Airbnb Price Prediction Modeling, Independent Developer

01/2026 – 04/2026

Tech Stack: Python, Pandas, Scikit-learn, RandomForest, ColumnTransformer, GridSearchCV

- Built a price prediction model on NYC Airbnb listings, executing a complete supervised learning pipeline from problem formulation, data splitting, and exploratory data analysis to feature engineering and model evaluation
- Designed a preprocessing strategy using ColumnTransformer to handle numerical and categorical features, including location encoding, room type transformation, and review-based feature construction
- Benchmarked RandomForest against DummyRegressor as a baseline, evaluating model performance through cross-validation, R^2 , and MSE; analyzed the trade-off between model complexity and generalization capability
- Performed hyperparameter tuning via GridSearchCV to optimize model configuration, and critically discussed error sources, data limitations, and the practical constraints of applying the model to real-world pricing scenarios

AI-Powered Stock Analysis & Visualization Dashboard, Independent Developer

08/2025 – 11/2025

Tech Stack: Python, AI-assisted Development, Streamlit, Data Visualization

- Designed an interactive A-share analysis dashboard for personal investment review, integrating market quotes, tick-by-tick data, capital flows, large-order behavior, and intraday price movements to support data-driven decision-making
- Structured data fetching, cleaning, metric calculation, and charting components through AI-assisted development, refactoring ad-hoc scripts into a maintainable and modular project architecture
- Implemented exploratory modules, including anomaly detection, short-term directional classification, and behavioral clustering, to distinguish observational indicators from actionable trading signals and investigate underlying data patterns

China Glass Futures Quantitative Trading Strategy, Independent Developer

09/2024 – 03/2025

Tech Stack: TradingView, PineScript, Python, Pandas, Matplotlib, Backtesting Analysis

- Developed a quantitative trading strategy for China glass futures, backtesting trade signals, trend phases, and risk control logic on historical market data to evaluate strategy validity across different market conditions
- Assessed strategy performance across different market regimes, focusing on risk-adjusted metrics including drawdown, trade frequency, and parameter stability, rather than relying on single-metric evaluations such as total return
- Integrated transaction costs, slippage, and parameter sensitivity into risk analysis, shifting the project emphasis from performance presentation to strategy interpretability and robustness

SKILLS

- **Language:** Mandarin (Native); English (Proficiency)
- **Programming Languages:** Python, Java, C++, R, JavaScript
- **Data Science & Machine Learning:** Scikit-learn, Pandas, NumPy, Matplotlib, Streamlit; regression and classification models, feature engineering, hyperparameter tuning, time-series forecasting, cross-validation
- **NLP & Unsupervised Learning:** LDA, TF-IDF, GloVe, K-Means, DBSCAN; text vectorization, topic modeling, semantic analysis
- **Tools & Core CS:** Git, Jupyter Notebook; Data Structures & Algorithms, OOP